Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**SEAMIIC: Autonomous parts handling and quality control** "Smart End-2-End Autonomous Manufacturing Innovation Industry 4.0 Capabilities, combining data & ai, intelligent imaging and (opto)mechatronics to reduce cost of handling of parts from post processing 3D printing or repair/reuse product streams" 			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine (√), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical/materials		*Electrical/Electronics*	
	Medium Category	Small Category	Medium Category	Small Category
	Robotics/Automation Machinery	*Assembly/Precision Transfer Technologies, Measurement/Sensor Technologies for Automation*	*Measuring Instruments*	*Measurement Sensors and Components*

Concept of Proposed Technology	*SEAMIIC (Smart, Efficient, Autonomous Manufacturing through Intelligent Imaging and Control) is a modular, AI-driven production platform for autonomous handling, inspection, and quality control of high-mix, low-volume parts in high-tech manufacturing. The focus is on post-processing of 3D-printed components and automated parts handling for repair/reuse/refurbishment, enabling cost effective manufacturing for high-mix-low-volume high tech products and components. SEAMIIC integrates advanced robotics, intelligent imaging (including ultrasound-based NDT), and data-driven decision-making to deliver flexible, scalable automation for SMEs to optimize handling cost for high mix – low volume – high complexity product portfolio.*
Rationale for Joint R&D Project	*The rapid adoption of additive manufacturing and refurbishment in high-tech industries creates new challenges in post-processing and quality assurance, especially for SMEs. Manual handling is labor-intensive and inconsistent. Joint R&D is needed to combine TNO's expertise in AI, imaging, and NDT with Korean SME capabilities in automation and manufacturing, accelerating the development and deployment of autonomous, high-quality post-processing solutions.*
Key Words	1. *Robotics & Autonomous handling parts & components* 2. *Autonomous Quality control Parts & component* 3. *Intelligent Imaging* 4. *3D vision* 5. *Integrated 3D nesting* 6. *Post processing additive manufacturing* 7. *Processing circular value streams repair, reuse, refurbish* 8. *Automated product flows*

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	*TRL 4 (Lab validation of integrated modules)*	End of the Project	*TRL 6 (Pilot demonstration in SME environment: Production & performance evaluation of prototypes at a pilot scale)*

R&D Objectives

Key R&D objectives

1. Develop an AI-driven platform for autonomous post-processing and quality control of 3D-printed and repair/reuse/refurbished parts for different use case applications.
2. Develop integrated modular productions cells for autonomous post-processing and quality control of 3D-printed and repair/reuse/refurbished parts.
3. Integrate intelligent imaging (3D vision, 3D nesting, ultrasound NDT) for in-line defect detection and process optimization.
4. Demonstrate and validate flexible robotic handling for diverse part geometries and materials.

SEAMIIC (Smart, Efficient, Autonomous Manufacturing through Intelligent Imaging and Control) will deliver a modular, AI-driven platform that **automates parts handling and in-line quality control** across post-processing of 3D-printed components and repair/reuse/refurbishment flows in **high-mix/low-volume** settings. The core objective is to combine **advanced robotics**, **intelligent imaging** (3D vision, 3D nesting, ultrasound-based NDT), and **data-driven decision-making** into configurable cells that achieve **zero-defect, low-touch material flow** while remaining cost-effective for SMEs.

To ensure industrial adoption, the R&D follows a **systems-engineering approach by use-case**: **WP1** elicits **User Requirement Specifications (URS)** per SME group scenario; **WP2** derives the **System Requirement Specifications (SRS)** and reference architecture; **WP3** implements **prototype cells** with integrated imaging/robotics/AI; **WP4** runs **test & validation** (factory acceptance in NL and, where required, **on-site validation in Korea**) while capturing **CAPEX/OPEX** implications (cycle time, staffing, energy, yield) to mature the **business case** for scale-out. This staged approach reflects best practice for maturing autonomous solutions from “islands of autonomy” toward **site-wide integration** that aligns **technology, ecosystem, and business model** maturity.

SEAMIIC is designed to **interoperate with DWI/Instant Instructions** so that knowledge capture during prototype operation (video, process telemetry, operator annotations) can be **transformed into step-wise, bilingual digital work instructions** for rapid onboarding, change management

	and quality assurance; this ensures human-centric deployment and sustainable knowledge retention alongside automation. **Important:** The **R&D objectives will be further detailed and prioritized based on the concrete use-cases of the participating Korean SMEs** (sector, part mix, takt constraints, legacy MES/QMS), ensuring localization of templates, UI/UX, safety concepts, and integration paths with Korean factory IT/OT.
R&D Target Specifications	**Key R&D target specifications are:** 1. Automated handling of parts with best available cycle time 2. In-line defect detection accuracy > 95% (3D vision + ultrasound) 3. Recognition & adaptability to at least 100 different part types without manual reprogramming 4. Reduction of manual labor by > 80% in post-processing/refurbishment lines 5. Overall Operational Equipment Efficiency of >90% 6. Scalable and modular prototype validated for several manufacturing applications 7. Cost reduction for handling and quality control of more than a factor 100. In more detail the following specifications w.r.t. the SEAMIIC testbed for parts handling and quality control can be further assessed based on Use cases from Korean SMEs: **1. Autonomous handling breadth & adaptability** • Grasping, orientation and transport of **≥100 distinct part types** (polymers/metals, variable surface finish) **without per-part reprogramming**, using part-family templates, 3D vision, and learned grasp strategies. • **Changeover ≤5 minutes** between part families via recipe selection and automatic sensor/robot re-configuration. 2. **In-line quality detection & decisioning** • **Defect detection accuracy ≥95%** at line speed by fusing **3D vision + ultrasound NDT** with rule-based and learning-based inference; automated routing to rework/accept/reject with provenance logging. 3. **Throughput, OEE, and labor reduction** • Achieve **best-available cycle time** for target use-cases with **OEE ≥90%** at pilot scale; demonstrate **≥80% reduction** in manual handling labor across post-processing/refurbishment lines while maintaining or improving first-time-right. 4. **Human-centric deployment with DWI integration** • Auto-generation of **bilingual (KR–EN) work instructions** from captured operations (video/CAD/sensor streams) to reduce onboarding time **≥30–40%** and support safe exceptions handling; closed-loop feedback from shop-floor back into instruction quality. 5. **Systems & data interoperability** • Interfaces to **≥2 Korean MES/QMS/ERP systems** using standardized connectors (e.g., OPC UA/AAS-style data models) to assure data lineage, auditability, and recipe control across autonomous cells. 6. **Deployment, safety, and business metrics** • On-prem, hybrid, or sovereign-cloud deployment modes with role-based access and traceability.

	• **Validated safety concept** for mixed human/robot operations per use-case, evolving from fenced cells toward **progressive autonomy** consistent with maturity models for autonomous solutions. • Business-case KPIs: **unit handling cost**, **energy per part**, **rework rate**, **yield uplift**, feeding into **CAPEX/OPEX** calculators to support go/no-go decisions for Korean SME implementation. **Note:** As with the objectives, the **final target specifications will be refined with Korean SME pilots**; validation will start on **TNO's NL testbed**, with the **option to ship modular test cells to Korea** for **on-site validation** and regulatory/user acceptance where needed. **Research Trajectory & Governance (WP1–WP4)** • **WP1 — URS (per use-case):** capture shop-floor workflows, takt time, part families, quality criteria, safety constraints, IT/OT context; define **validation KPIs** and business metrics up front. This responds to the well-documented need to anchor autonomy in **ecosystem and business model realities**, not only in technology. • **WP2 — SRS & Architecture:** derive functional/non-functional specs; select imaging stack (3D vision/ultrasound), robot and gripper kits, edge inference, and data models (incl. AAS-style semantics) to ensure interoperability and upgradability. • **WP3 — Prototype (NL):** integrate cells; enable **DWI/Instant Instructions** coupling for real-time knowledge capture and instruction generation; execute FAT on throughput/OEE/quality. • **WP4 — Test & Validation (NL → Korea):** run tests at TNO and **Korean SME sites** (by shipping the modular testbed where required); quantify **CAPEX/OPEX** and refine the commercialization roadmap (service model, spares, training, SLAs), consistent with maturity frameworks for autonomous operations.
R&D Plan	**High level planning to be updated based on Korean SME use cases:** • **Month 0-6:** Define User Requirement Specification and System requirement specification based on SME use-cases. • **Month 6-18:** develop and extend existing lab prototype with System architecture, AI model development, imaging/NDT module integration, robotic handling integration incl. preliminary test runs • **Month 18-30:** Pilot demonstration, performance validation, • **Month > 30:** technology & commercialization roadmap development and industrial implementation at participating SME's

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Lead AI and data analytics development for autonomous decision-making • Develop and integrate intelligent imaging modules (vision, ultrasound/DVM NDT) • Develop modular test modules for advanced manufacturing: advanced handling, advanced quality control, advanced identification, etc. • Provide prototype test bed for testing and validation of efficiency improvements • System integration and validation in pilot environments • Knowledge transfer and training for SME partners
Expected R&R of Korean SME Partner	• Provide manufacturing use cases and requirements • Develop and adapt robotic handling hardware • Support pilot implementation and validation in operational settings • Contribute to commercialization and market introduction in Korea

5. Expected Outcomes/Impact

Expected Outcomes and Impact	• **Technological:** First-of-its-kind autonomous post-processing and quality control platform for high-mix, high-tech manufacturing • **Commercial:** Accelerated adoption of 3D printing and repair/reuse/refurbishment by SMEs, increasing productivity, lowering manufacturing cost and boosting circular manufacturing and new business models • **Economic:** Increased productivity, reduced labor costs, improved product quality, and enhanced competitiveness for Korean and European SMEs

6. References

1. **3D Nesting – TNO**
   https://www.tno.nl/en/digital/smart-manufacturing/manufacturing-throughput-quality/3d-nesting-quality-control-manufacturing/
   [tno.nl]
2. **DVM NDT (Direct Velocity Mapping) – TNO**
   https://www.tno.nl/en/digital/smart-manufacturing/manufacturing-throughput-quality/non-destructive-inspection/
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3. **SEAMIIc: Autonomous Parts Handling & Quality Control – TNO**
   https://www.tno.nl/en/digital/smart-manufacturing/manufacturing-throughput-quality/seamiic-autonomous-handling-quality/
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4. **Flexible / Autonomous Manufacturing – TNO**
   https://www.tno.nl/en/digital/smart-manufacturing/flexible-manufacturing/
   [tno.nl]
5. **AAS & RDF Modelling in Industry 4.0 – TNO**
   https://publications.tno.nl/publication/34641154/LyQGGI/rongen-2023-modelling.pdf
   [publications.tno.nl]
6. **Patents / IP**

   - PLT2020075 Non-contact inspection technique using MEMs-microphones
   - PLT2023031 Non-contact inspection technique for anisotropic composites
   - PLT2023046 Patent application 3D nesting
   - I-Depot visual robot feedback for tracking and picking
   - PLT2017057 User assistance in assembly – operator work station